

Step 1 — Imaging model

observe a discrete image on K pixels ($K = 25 \times 25 = 625$ in my code). For a single emitter at position $\theta = (x, y, z)$ with known photon count N :

Expected signal in pixel k :

$$\mu_k(\theta) = N \cdot q_k(\theta)$$

where q_k is the pixel-integrated, normalized PSF fraction ($\sum_k q_k = 1$). Observed counts Y_k follow Poisson statistics with effective background β (Step 4).

Code: `crlb/fisher_crlb.py` — $\mu_k = N \cdot q_k$ via `_mu_from_q()`;

Step 2 — Pixel-integrated PSF models

2A. Astigmatic PSF

In-focus widths σ_{x0}, σ_{y0} with astigmatism parameters c, d (nm):

$$\sigma_x(z) = \sigma_{x0} \cdot \sqrt{1 + ((z+c)/d)^2 + a_x \cdot ((z+c)/d)^3 + b_x \cdot ((z+c)/d)^4}$$

$$\sigma_y(z) = \sigma_{y0} \cdot \sqrt{1 + ((z-c)/d)^2 + a_y \cdot ((z-c)/d)^3 + b_y \cdot ((z-c)/d)^4}$$

JBO 2013 preset values:

Parameter	Value
σ_{x0}, σ_{y0}	140, 135 nm
c, d	50, 500 nm
Pixel size	128 nm
N photons	3000

2B. Double-helix PSF

Lobe rotation angle:

$$\varphi(z) = \varphi_{\max} \cdot (z / z_{\text{ref}})$$

Two Gaussian lobes centered at:

$$(x_1, y_1) = (x + (s/2)\cos \varphi, y + (s/2)\sin \varphi)$$

$$(x_2, y_2) = (x - (s/2)\cos \varphi, y - (s/2)\sin \varphi)$$

Combined normalized PSF:

$$q_k = (1/2)(q_k^{(1)} + q_k^{(2)}), \quad \sum_k q_k = 1$$

This is a simplified rotating double-lobe model (Badieirostami 2010 style), not a full phase-mask wave-optics PSF.

Parameter	Value (JBO preset)
Lobe σ	115 nm
Lobe separation s	480 nm
ϕ max	π rad
z ref	600 nm

Step 3 — Fisher information matrix (Poisson pixels)

For independent Poisson pixels, the log-likelihood is

$$\ell(\theta) = \sum_k [Y_k \log(\mu_k + \beta) - (\mu_k + \beta) - \log(Y_k!)].$$

The Fisher information matrix is:

$$I_{ij}(\theta) = \sum_k [1 / (\mu_k(\theta) + \beta)] \cdot (\partial \mu_k / \partial \theta_i)(\partial \mu_k / \partial \theta_j)$$

Step 4 — Effective background

For the JBO 2013 preset I used your paper combined Poisson + Gaussian readout noise model.

The effective background per pixel in the Fisher denominator is:

$$\beta_{\text{eff}} = N \cdot (1/\gamma_p + 1/\gamma_g)$$

$$\gamma_p = 400000 \text{ nm}^2/\text{emitter} \text{ (SPNR} = 0.4 \text{ } \mu\text{m}^2/\text{emt}),$$

$$\gamma_g = 600000 \text{ nm}^2/\text{emitter} \text{ (SGNR} = 0.6 \text{ } \mu\text{m}^2/\text{emt}),$$

$$N = 3000 \rightarrow \beta_{\text{eff}} = 0.0125 \text{ photons/pixel.}$$

Step 5 — Numerical computation of derivatives

Because $q_k(x, y, z)$ involves pixel integration and (for DH-PSF) lobe rotation, i did not use closed-form $\partial q / \partial \theta$. Instead, central finite differences in `fisher_information()`:

$$\partial q_k / \partial x \approx [q_k(x+\Delta x, y, z) - q_k(x-\Delta x, y, z)] / (2\Delta x)$$

Default steps in code: $\Delta x = \Delta y = 1 \text{ nm}$, $\Delta z = 5 \text{ nm}$.

Then $\partial \mu_k / \partial \theta = N \cdot \partial q_k / \partial \theta$, and the FIM sum runs over all $K = 625$ pixels.

Step 6 — Cramér–Rao lower bound (CRLB)

For unbiased estimators, $\text{Var}(\theta_i) \geq [I^{-1}(\theta)]_{ii}$. Localization precision limits:

$$\sigma_x^{\text{CRLB}} = \sqrt{[I^{-1}]_{11}}, \quad \sigma_y^{\text{CRLB}} = \sqrt{[I^{-1}]_{22}}, \quad \sigma_z^{\text{CRLB}} = \sqrt{[I^{-1}]_{33}}$$

Code: `crlb_from_fisher()` and `crlb_xyz_from_fisher()`

Step 7 — Numerical example ($z = 0$ nm)

The following matrices were computed by running `fisher_information()` at $(x,y,z) = (0,0,0)$ with the JBO 2013 preset

Astigmatic PSF — FIM at $z = 0$ (units: nm^{-2}):

row/col	$\partial/\partial x$	$\partial/\partial y$	$\partial/\partial z$
0.141514	-0.000000	-0.000000	I-xz
-0.000000	0.151451	-0.000000	I-yz
-0.000000	-0.000000	0.000408	I-zz

CRLB: $\sigma_x = 2.658$ nm, $\sigma_y = 2.570$ nm, $\sigma_z = 49.496$ nm. The small I_{33} ($\sim 4 \times 10^{-4}$) reflects weak z-encoding at focus ($\sigma_x \approx \sigma_y$).

Double-helix PSF — FIM at $z = 0$ (units: nm^{-2}):

row/col	$\partial/\partial x$	$\partial/\partial y$	$\partial/\partial z$
0.151282	-0.000000	-0.000000	I-xz
-0.000000	0.205209	-0.000000	I-yz
-0.000000	-0.000000	0.302513	I-zz

CRLB: $\sigma_x = 2.571$ nm, $\sigma_y = 2.208$ nm, $\sigma_z = 1.818$ nm. At focus, DH-PSF axial precision (1.8 nm) is far better than astigmatic (49.5 nm).

Step 8 — Extensions

8A. Unknown intensity I

If intensity is unknown, $\theta = (x, y, z, I)$ with $\mu_k = I \cdot q_k$. Then $\partial \mu_k / \partial I = q_k$ and the FIM is 4×4 . Position CRLB increases compared to known N .

8B. Multi-emitter

For M emitters, $Q_k = \sum_m q_k^{(m)}(\theta_m)$. Derivatives of emitter m use the summed mean image.